University of Glasgow

Friday 09 May 2025
14:00-15:30 BST
Duration: 1 hour 30 minutes
Timed exam – fixed start time

DEGREE OF MSc

Deep Learning for MSc
COMPSCI5103

(Answer All Questions)

This examination paper is an open book, online assessment and is worth a total of 60 marks.

1. You should answer the following questions by examining the following PyTorch model.

```
class Net2d(nn.Module):
    def __init__(self):
        super(Net2d, self).__init__()

        self.fc1 = nn.Linear(2, 10)
        self.fc2 = nn.Linear(10, 10)
        self.fc3 = nn.Linear(10, 1)

    def forward(self, x):
        x = F.relu(self.fc1(x))
        x = F.relu(self.fc2(x))
        x = torch.sigmoid(self.fc3(x))
        return x
```

(a) Describe what type of input this network takes and the type of output it produces. What machine learning task does it do? [3]

(b) What type of PyTorch loss function should be used for this network? [1]

(c) Calculate how many parameters *each layer of the network* contains. [3]

(d) The following optimizer is used for the above model:

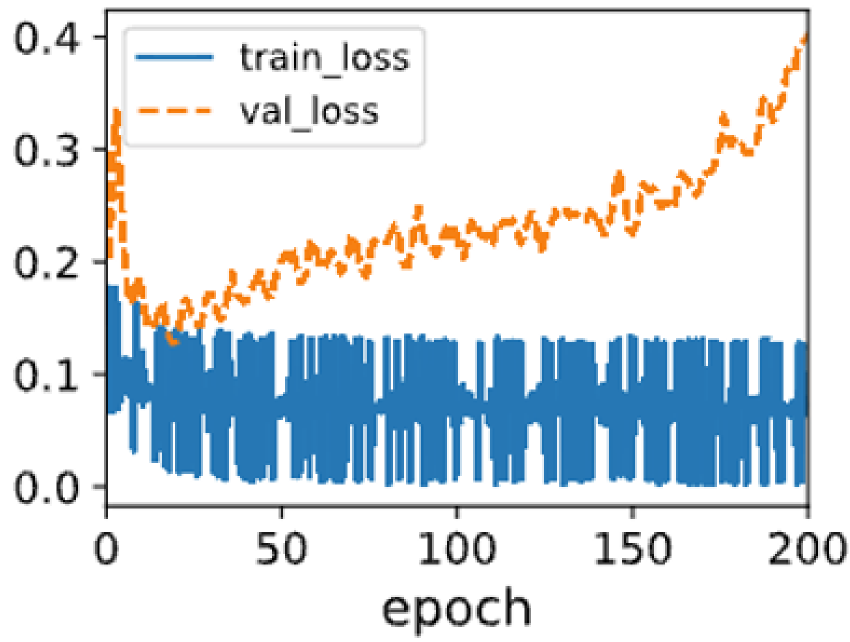
```
optimizer = optim.SGD(net2d.parameters(), lr=0.01,
                      momentum=0.9, weight_decay=0.001)
```

(i) Explain what would happen in terms of the loss function over time if too large a value for `lr` was chosen. Also explain what would happen if too small value was chosen. [2]

(ii) Explain what the `momentum` term does and what sort of loss surface it may help with. [3]

(iii) Explain what the `weight_decay` term does and what it may help with. [3]

(e) This model produced the loss curve given below. Describe what this loss curve indicates and suggest *four specific different ways* to help resolve this issue. [5]



2. **(a)** For answering these questions, you should write out matrices in ASCII format, for example:

```
1, 2
3, 4
```

This question part concerns the following 2D kernel:

-1	-1	-1
+1	+1	+1

This is applied to the following 2D image:

0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

(i) What type of image features would you expect this kernel to detect? [1]

(ii) Calculate the output tensor if the kernel was applied to the image with padding = 0, stride = 1 ? [3]

(iii) Calculate the output tensor if the kernel was applied to the image with padding = 1, stride = 2 ? [4]

(iv) Calculate the output tensor if maxpooling with size (5 rows,1 column) and stride = 2 was applied to the image above? [2]

(b) The following questions relate to the GAN architecture used for image generation.

(i) Describe the two architectural parts of the GAN architecture and the purpose of each part when generating images.

[4]

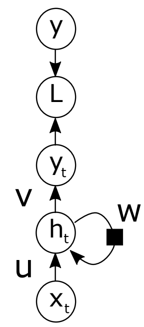
(ii) The equation below appears as Equation 1 in the Goodfellow et al. (2014) paper about GANs. For this equation, describe the meaning of the following terms:
`G, D, x ~ p_data(x), D(x), z ~ p_z(z), D(G(z)).`

[6]

$$\min_G \max_D V(D,G) = \mathbb{E}_{\boldsymbol{x} \sim p_{\text{data}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{z} \sim p_{\boldsymbol{z}}(\boldsymbol{z})}[\log(1 - D(G(\boldsymbol{z})))].$$

3. **(a)** For answering these questions, you should write out equations in ASCII format, for example:
`dy/dw_1` for $\frac{\partial y}{\partial w_1}$, etc.

The following figure shows the structure of a vanilla RNN that uses scalar values.



It employs the following formulae:

$$\begin{aligned} h_t &= ReLU(wh_{t-1} + ux_t) \\ y_t &= c + vh_t \\ L &= \frac{1}{2}(y - y_t)^2 \end{aligned}$$

(i) Assume u = -3.0, v = 3.0, w = 0.5, c = +1.0 (and $h_{-1} = 0.0$). Two of these recurrent units are joined together. Do a 'forward pass' of the network using the sequence of input values $x_0 = -1.0$, $x_1 = +2.0$. Calculate the sequence of outputs for y_t. Show your working. [5]

(ii) Determine **the expressions (containing variables)** for $\frac{\partial y_0}{\partial x_0}$ for both when the input to the ReLU is less than or equal to zero, and for when its input is greater than zero. [5]

(b) What are the advantages and limitations of the Transformer architecture compared to an RNN architecture for processing text? [4]

(c) Consider a Transformer block with an input sequence of $l = 1024$ tokens, each embedded in a space of dimension $d_e = 100$. Let us assume that the self-attention module block has 2 attention heads and that the key and query dimensions are both $d_k = 10$, the value dimension is $d_v = 20$ and the output dimension is $d_o = 100$. How many parameters would the self-attention module require? Explain your reasoning detailing each variable.

[6]

END OF QUESTION PAPER